Digital Mohalla — Class diagram, Student 1 packages (design step A7, v1.0): Module 1 accounts & locality, Module 3 borrowing lifecycle, Module 5 indirect trust

Python design classes (iteration 1). Each class lists the FRs it realises. Student 2 classes appear only as the interfaces Student 1 depends on (orange).

accounts (Module 1)

User
id: UUID
display_name: str
verification_level: VerificationLevel
platform_role: PlatformRole
status: AccountStatus
is_synthetic: bool
FR-A-04 (part), NFR-S-07: no CNIC / address fields

1 — 0..*

Membership
user_id: UUID
community_id: UUID
role: MembershipRole
status: MembershipStatus
is_moderator_of(community_id): bool
FR-A-09 (seeded), NFR-S-03

1 — 0..*

«value»
LocalityResult
membership_id: UUID
method: VerificationMethod
outcome: LocalityOutcome # never negative
geofence_signal: GeofenceSignal | None
invitation_id: UUID | None
permission_refused, mock_provider_detected, duplicate_suspected: bool
approved_by: UUID | None # moderator path
checked_at, valid_until: datetime
FR-A-07, FR-A-08, NFR-PR-02 (no coordinates)

uses — 0..1 — 0..1

Invitation
community_id, inviter_id: UUID
code_hash: bytes # SHA-256 only
expires_at: datetime
FR-A-07, UC-A3 3d (A11; single use)

«use»

«service»
GeofenceInvitationVerifier
evaluate(fix, permission_refused, invitation, duplicate_suspected)
: LocalityResult # fix used in memory only
FR-A-07, FR-A-08 (A11 decision table)

«interface»
LocalityChecker
outcome_for(user_id, community_id): LocalityOutcome
FR-B-07 (UC-B5 step 4)

StoredResultLocalityChecker
outcome_for(...): LocalityOutcome # latest valid stored result
FR-B-07, FR-A-08; reads results written by the verifier

AuthService
login(identifier, password): Session
current_actor(token): Actor
FR-A-04 (part), NFR-S-01

lifecycle (Module 3)

«application service»
BorrowingService
- store: EventStore
- validator: TransitionValidator
- engine: AssuranceEngine
- locality: LocalityChecker
- outcomes: OutcomeSink
create_request(cmd: CreateRequest): BorrowingTransaction
accept(id, actor) / decline(id, actor, reason)
cancel(id, actor)
record_handover_checklist(id, actor, report)
confirm_handover(id, actor, agreed)
initiate_return(id, actor)
confirm_return(id, actor, report)
open_dispute(id, actor, reason) / withdraw_dispute(id, actor)
apply_decision(id, decision: ModeratorDecision)
FR-B-05, B-06, B-07, B-10, B-11, B-13, B-15, B-16, B-17

«create, load»

«aggregate»
BorrowingTransaction
id: UUID # = stream_id
item_id, borrower_id, lender_id: UUID
requested_period: DateTimeRange
lifecycle_state: LifecycleState
hold_reason: HoldReason | None
due_at: datetime | None
stream_version: int
apply(event: LifecycleEvent): None
replay(events): BorrowingTransaction «static»
FR-B-08, FR-B-09, NFR-R-03

1 — {ordered} — 1..*

«value, immutable»
LifecycleEvent
event_id: int
stream_id: UUID
stream_seq: int
event_type: EventType
schema_version: int
actor_id: UUID | None
occurred_at: datetime
previous_state, new_state: LifecycleState
evidence_refs: dict
authorisation: str
config_version: str | None
payload: dict # ADR-0005
FR-B-09, FR-N-03, NFR-D-01 (frozen schema)

«use»

TransitionValidator
- table: TransitionTable
validate(txn, event_type, target, actor): Transition
raises InvalidTransition, NotAuthorised, GuardFailed
FR-B-08, NFR-R-02, NFR-S-02, NFR-S-03

«use»

TransitionTable
rows: list[Transition] # 34, from transition_table.csv
lookup(state, event_type, target): Transition | None
FR-B-08, NFR-R-02 (A3 table = test matrix)

1 — 34

«value»
Transition
from_state, to_state: LifecycleState
event_type: EventType
allowed_roles: frozenset[ActorRole]
guard: Callable[[Context], bool]
ref: str # T01..T26
FR-B-08

«stores»

«interface»
EventStore
append(stream_id, expected_seq, event): None
raises StaleStream
load(stream_id): list[LifecycleEvent]
FR-B-09, ADR-0002 (unique stream_id + stream_seq)

PostgresEventStore
append(...) # one DB transaction: event + booking
load(...)
FR-M-05, NFR-S-05 (append-only, hash chain)

Booking
transaction_id, item_id: UUID
period: DateTimeRange
status: BookingStatus {HELD, RELEASED}
FR-B-05, NFR-R-01 (EXCLUDE constraint, ADR-0002)

0..1

«value»
ConditionReport
phase: ConditionPhase
reporter_role: PartyRole
checklist: dict
condition_agreed: bool
deposit_received: bool | None
reduced_evidence: bool
FR-B-11, FR-B-13; I-4 record-only deposit

0..*

«enumeration»
Lifecycle State / EventType / HoldReason
14 states, 21 event types, 5 hold reasons
(lists: State_Machine_Design.md v2.0)
FR-B-08, FR-B-09, ADR-0003

«value»
Actor
user_id: UUID | None
role: ActorRole
authorisation: str
NFR-S-02

«use» assess()

«use»

assurance.indirect (Module 5, S1 part)

TrustEdgeProjector
on_outcomes(outcomes: list[Outcome]): None
FR-T-04 (edges from verified outcomes only, FR-T-03)

«upserts»

TrustEdge
source_id, target_id: UUID
role: PartyRole
metric: OutcomeMetric
weight: float # Beta posterior mean, ADR-0006
evidence_count: int
FR-T-04

0..* — 1

TrustGraph
- edges: dict[UUID, list[TrustEdge]]
neighbours(u, role, metric): list[TrustEdge]
FR-T-04

«use»

KHopPathSearch
k: int # from config (initial 3)
paths(u, v, role, metric): list[Path] # simple paths, len <= k
FR-T-04, NFR-P-03 (k sweep)

«creates»

«use»

«value»
Path
nodes: list[UUID]
hops: int
q: float
FR-T-04

1..*

«value»
IndirectEvidence
paths: list[Path]
strength_i: float
rho: float # <= 0.6
t_indirect: float
prior_backed: bool
FR-T-04, FR-T-01 (paths in explanation, ES-13)

«returns»

IndirectTrustEngine
gamma, cap: float # config
indirect_evidence(lender, borrower, role, metric, t_v, b): IndirectEvidence
FR-T-04 (realises IIndirectEvidence)

assurance (Module 5, Student 2) — boundary only

«interface»
AssuranceEngine (IAssess)
assess(request: AssessmentRequest): AssessmentResult
FR-T-01..T-07; implemented by Student 2

«returns»

«value»
AssessmentResult
assurance_state: AssuranceState
policy_decision
rule_fired: str
hold_reason | None
config_version: str
FR-T-01, NFR-D-01

«interface»
OutcomeSink
record_outcome(closing_event_id)
FR-B-15

Notes
• lifecycle_state (where the loan is) and AssessmentResult.assurance_state (what the evidence says) are separate attributes of separate classes (ES-04: HELD_FOR_REVIEW with REVIEW_RECOMMENDED).
• BorrowingTransaction is rebuilt only through apply(); nothing sets lifecycle_state directly (FR-B-09).
• Student 2 implements AssuranceEngine and OutcomeSink; Student 1 implements IndirectTrustEngine, which the AssuranceEngine calls through IIndirectEvidence.
• Module 9 (Student 3) calls BorrowingService.apply_decision() through ILifecycleCommands; it is not drawn here.
• Relationship ends: ◆ composition, ◇ aggregation, dashed open arrow = dependency, dashed hollow triangle = realisation.